

Appendices

Appendix A: Model 1 - Gemini RDF/OWL Code (Gemini_KG.ttl)

```
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .

@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

@prefix owl: <http://www.w3.org/2002/07/owl#> .

@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

@prefix ex: <http://example.org/streaming#> .

<http://example.org/streaming> a owl:Ontology ;

    rdfs:label "Gemini Streaming Impact Ontology" .

# --- Classes ---

ex:MediaPlatform a owl:Class .

ex:StreamingService a owl:Class ; rdfs:subClassOf ex:MediaPlatform .

ex:GlobalStreamingService a owl:Class ; rdfs:subClassOf ex:StreamingService
.

ex:LocalStreamingService a owl:Class ; rdfs:subClassOf ex:StreamingService
.

ex:TraditionalOutlet a owl:Class ; rdfs:subClassOf ex:MediaPlatform .

ex:CommercialTVChannel a owl:Class ; rdfs:subClassOf ex:TraditionalOutlet .

ex:PublicBroadcaster a owl:Class ; rdfs:subClassOf ex:TraditionalOutlet .

ex:CinemaChain a owl:Class ; rdfs:subClassOf ex:TraditionalOutlet .

ex:ContentItem a owl:Class .

ex:OriginalProduction a owl:Class ; rdfs:subClassOf ex:ContentItem .

ex:LocalOriginalSeries a owl:Class ; rdfs:subClassOf ex:OriginalProduction
.

ex:GlobalOriginalFilm a owl:Class ; rdfs:subClassOf ex:OriginalProduction .

ex:LicensedContent a owl:Class ; rdfs:subClassOf ex:ContentItem .

ex:AcquiredDramaSeries a owl:Class ; rdfs:subClassOf ex:LicensedContent .

ex:LicensedDocumentary a owl:Class ; rdfs:subClassOf ex:LicensedContent .
```

```

ex:DemographicGroup a owl:Class .

ex:AgeCohort a owl:Class ; rdfs:subClassOf ex:DemographicGroup .

ex:YouthCohort a owl:Class ; rdfs:subClassOf ex:AgeCohort .

ex:AdultCohort a owl:Class ; rdfs:subClassOf ex:AgeCohort .

ex:GeographicRegion a owl:Class ; rdfs:subClassOf ex:DemographicGroup .

ex:MiddleEastRegion a owl:Class ; rdfs:subClassOf ex:GeographicRegion .

ex:EuropeanRegion a owl:Class ; rdfs:subClassOf ex:GeographicRegion .

ex:SocioBehavioralImpact a owl:Class .

ex:PersonalBehavioralImpact a owl:Class ; rdfs:subClassOf
ex:SocioBehavioralImpact .

ex:SleepDisruption a owl:Class ; rdfs:subClassOf
ex:PersonalBehavioralImpact .

ex:SocialIsolation a owl:Class ; rdfs:subClassOf
ex:PersonalBehavioralImpact .

ex:PublicCulturalShift a owl:Class ; rdfs:subClassOf
ex:SocioBehavioralImpact .

ex:DeclineInReading a owl:Class ; rdfs:subClassOf ex:PublicCulturalShift .

ex:CinemaAbandonment a owl:Class ; rdfs:subClassOf ex:PublicCulturalShift .

ex:GeopoliticalFactor a owl:Class .

ex:RegulatoryPolicy a owl:Class ; rdfs:subClassOf ex:GeopoliticalFactor .

ex:StateCensorship a owl:Class ; rdfs:subClassOf ex:RegulatoryPolicy .

ex:LocalContentQuota a owl:Class ; rdfs:subClassOf ex:RegulatoryPolicy .

ex:CulturalConflict a owl:Class ; rdfs:subClassOf ex:GeopoliticalFactor .

ex:DiplomaticTension a owl:Class ; rdfs:subClassOf ex:CulturalConflict .

# --- Object Properties ---

ex:producesContent a owl:ObjectProperty ; rdfs:domain ex:MediaPlatform ;
rdfs:range ex:ContentItem .

ex:isProducedBy a owl:ObjectProperty ; rdfs:domain ex:ContentItem ;
rdfs:range ex:MediaPlatform .

ex:targetsDemographic a owl:ObjectProperty ; rdfs:domain ex:ContentItem ;
rdfs:range ex:DemographicGroup .

```

```

ex:isWatchedBy a owl:ObjectProperty ; rdfs:domain ex:ContentItem ;
rdfs:range ex:DemographicGroup .

ex:causesImpact a owl:ObjectProperty ; rdfs:domain ex:MediaPlatform ;
rdfs:range ex:SocioBehavioralImpact .

ex:experiencesImpact a owl:ObjectProperty ; rdfs:domain ex:DemographicGroup ;
rdfs:range ex:SocioBehavioralImpact .

ex:isSubjectToPolicy a owl:ObjectProperty ; rdfs:domain ex:MediaPlatform ;
rdfs:range ex:RegulatoryPolicy .

ex:enforcesPolicy a owl:ObjectProperty ; rdfs:domain ex:GeographicRegion ;
rdfs:range ex:RegulatoryPolicy .

ex:restrictsContent a owl:ObjectProperty ; rdfs:domain ex:StateCensorship ;
rdfs:range ex:ContentItem .

ex:mandatesQuotaFor a owl:ObjectProperty ; rdfs:domain ex:LocalContentQuota ;
rdfs:range ex:StreamingService .

ex:triggersConflict a owl:ObjectProperty ; rdfs:domain ex:ContentItem ;
rdfs:range ex:CulturalConflict .

ex:competesWith a owl:ObjectProperty ; rdfs:domain ex:StreamingService ;
rdfs:range ex:TraditionalOutlet .

ex:replacesActivity a owl:ObjectProperty ; rdfs:domain
ex:PersonalBehavioralImpact ; rdfs:range ex:PublicCulturalShift .

ex:isDistributedIn a owl:ObjectProperty ; rdfs:domain ex:ContentItem ;
rdfs:range ex:GeographicRegion .

ex:influencesOpinionOf a owl:ObjectProperty ; rdfs:domain ex:ContentItem ;
rdfs:range ex:DemographicGroup .

ex:opposesPlatform a owl:ObjectProperty ; rdfs:domain ex:CulturalConflict ;
rdfs:range ex:StreamingService .

ex:supportsPlatform a owl:ObjectProperty ; rdfs:domain ex:CulturalConflict ;
rdfs:range ex:StreamingService .

ex:originatesFrom a owl:ObjectProperty ; rdfs:domain ex:ContentItem ;
rdfs:range ex:GeographicRegion .

ex:correlatesWith a owl:ObjectProperty ; rdfs:domain ex:SleepDisruption ;
rdfs:range ex:SocialIsolation .

ex:adaptsStoryFrom a owl:ObjectProperty ; rdfs:domain ex:OriginalProduction ;
rdfs:range ex:ContentItem .

# --- Data Properties ---

ex:title a owl:DatatypeProperty ; rdfs:domain ex:ContentItem ; rdfs:range
xsd:string .

```

`ex:releaseYear a owl:DatatypeProperty ; rdfs:domain ex:ContentItem ;
 rdfs:range xsd:integer .`

`ex:durationMinutes a owl:DatatypeProperty ; rdfs:domain ex:ContentItem ;
 rdfs:range xsd:integer .`

`ex:primaryLanguage a owl:DatatypeProperty ; rdfs:domain ex:ContentItem ;
 rdfs:range xsd:string .`

`ex:imdbRating a owl:DatatypeProperty ; rdfs:domain ex:ContentItem ;
 rdfs:range xsd:float .`

`ex:isOriginal a owl:DatatypeProperty ; rdfs:domain ex:ContentItem ;
 rdfs:range xsd:boolean .`

`ex:episodesCount a owl:DatatypeProperty ; rdfs:domain
 ex:LocalOriginalSeries ; rdfs:range xsd:integer .`

`ex:seasonsCount a owl:DatatypeProperty ; rdfs:domain ex:LocalOriginalSeries
 ; rdfs:range xsd:integer .`

`ex:platformName a owl:DatatypeProperty ; rdfs:domain ex:MediaPlatform ;
 rdfs:range xsd:string .`

`ex:launchYear a owl:DatatypeProperty ; rdfs:domain ex:MediaPlatform ;
 rdfs:range xsd:integer .`

`ex:monthlySubscriptionFeeUSD a owl:DatatypeProperty ; rdfs:domain
 ex:StreamingService ; rdfs:range xsd:float .`

`ex:subscriberCountMillions a owl:DatatypeProperty ; rdfs:domain
 ex:StreamingService ; rdfs:range xsd:float .`

`ex:minAgeRequirement a owl:DatatypeProperty ; rdfs:domain ex:YouthCohort ;
 rdfs:range xsd:integer .`

`ex:maxAgeRequirement a owl:DatatypeProperty ; rdfs:domain ex:YouthCohort ;
 rdfs:range xsd:integer .`

`ex:populationSize a owl:DatatypeProperty ; rdfs:domain ex:GeographicRegion
 ; rdfs:range xsd:long .`

`ex:isContentBlocked a owl:DatatypeProperty ; rdfs:domain ex:StateCensorship
 ; rdfs:range xsd:boolean .`

`ex:requiredQuotaPercentage a owl:DatatypeProperty ; rdfs:domain
 ex:LocalContentQuota ; rdfs:range xsd:float .`

`ex:severityScale a owl:DatatypeProperty ; rdfs:domain ex:SleepDisruption ;
 rdfs:range xsd:integer .`

`ex:dailyAverageHoursSpent a owl:DatatypeProperty ; rdfs:domain
 ex:PersonalBehavioralImpact ; rdfs:range xsd:float .`

`ex:annualTicketSalesDropPercent a owl:DatatypeProperty ; rdfs:domain
 ex:CinemaAbandonment ; rdfs:range xsd:float .`

```

# --- Instances & Triples ---

ex:Netflix a ex:GlobalStreamingService .

ex:YesPlus a ex:LocalStreamingService .

ex:Channel12_Israel a ex:CommercialTVChannel .

ex:CinemaCity a ex:CinemaChain .

ex:Fauda_Series a ex:LocalOriginalSeries .

ex:SquidGame_Series a ex:GlobalOriginalFilm .

ex:Friends_Series a ex:AcquiredDramaSeries .

ex:TheSocialDilemma_Doc a ex:LicensedDocumentary .

ex:Youth_Cohort_18_25 a ex:YouthCohort .

ex:Adults_Cohort_50_Plus a ex:AdultCohort .

ex:Middle_East_Region a ex:MiddleEastRegion .

ex:European_Union_Region a ex:EuropeanRegion .

ex:Binge_Sleep_Loss a ex:SleepDisruption .

ex:Teen_Isolation a ex:SocialIsolation .

ex:Youth_Reading_Drop a ex:DeclineInReading .

ex:Movie_Theater_Decline a ex:CinemaAbandonment .

ex:Turkish_State_Censorship a ex:StateCensorship .

ex:EU_30Percent_Quota_Directive a ex:LocalContentQuota .

ex:Diplomatic_Row_Fauda a ex:DiplomaticTension .

ex:Scorsese_Cinema_Debate a ex:CulturalConflict .

ex:Netflix ex:producesContent ex:Fauda_Series .

ex:Netflix ex:producesContent ex:SquidGame_Series .

ex:Netflix ex:competesWith ex:Channel12_Israel .

ex:Netflix ex:competesWith ex:CinemaCity .

ex:Fauda_Series ex:originatesFrom ex:Middle_East_Region .

ex:Fauda_Series ex:isWatchedBy ex:Youth_Cohort_18_25 .

```

```

ex:Fauda_Series ex:triggersConflict ex:Diplomatic_Row_Fauda .
ex:Turkish_State_Censorship ex:restrictsContent ex:Fauda_Series .
ex:Middle_East_Region ex:enforcesPolicy ex:Turkish_State_Censorship .
ex:European_Union_Region ex:enforcesPolicy ex:EU_30Percent_Quota_Directive
.
ex:EU_30Percent_Quota_Directive ex:mandatesQuotaFor ex:Netflix .
ex:SquidGame_Series ex:targetsDemographic ex:Youth_Cohort_18_25 .
ex:Netflix ex:causesImpact ex:Binge_Sleep_Loss .
ex:Youth_Cohort_18_25 ex:experiencesImpact ex:Binge_Sleep_Loss .
ex:Binge_Sleep_Loss ex:correlatesWith ex:Teen_Isolation .
ex:Youth_Cohort_18_25 ex:experiencesImpact ex:Youth_Reading_Drop .
ex:Netflix ex:causesImpact ex:Movie_Theater_Decline .
ex:Movie_Theater_Decline ex:triggersConflict ex:Scorsese_Cinema_Debate .
ex:Scorsese_Cinema_Debate ex:opposesPlatform ex:Netflix .
ex:YesPlus ex:isDistributedIn ex:Middle_East_Region .

```

Appendix B: Model 2 - Perplexity / Prophecy RDF/OWL Code (Perplexity_KG.ttl)

```

@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix px: <http://example.org/perplexity#> .
<http://example.org/perplexity> a owl:Ontology ;
    rdfs:label "Perplexity Streaming Governance & Tech Ontology" .
# --- Classes ---
px:MediaPlatform a owl:Class .
px:StreamingPlatform a owl:Class ; rdfs:subClassOf px:MediaPlatform .

```

```

px:SubscriptionStreamingPlatform a owl:Class ; rdfs:subClassOf
px:StreamingPlatform .

px:AdSupportedStreamingPlatform a owl:Class ; rdfs:subClassOf
px:StreamingPlatform .

px:TraditionalMediaOutlet a owl:Class ; rdfs:subClassOf px:MediaPlatform .

px:BroadcastTelevisionChannel a owl:Class ; rdfs:subClassOf
px:TraditionalMediaOutlet .

px:PrintNewspaper a owl:Class ; rdfs:subClassOf px:TraditionalMediaOutlet .

px:RadioStation a owl:Class ; rdfs:subClassOf px:TraditionalMediaOutlet .

px:MediaContent a owl:Class .

px:EntertainmentContent a owl:Class ; rdfs:subClassOf px:MediaContent .

px:StreamingOriginalSeries a owl:Class ; rdfs:subClassOf
px:EntertainmentContent .

px:LicensedFilm a owl:Class ; rdfs:subClassOf px:EntertainmentContent .

px:InformationContent a owl:Class ; rdfs:subClassOf px:MediaContent .

px:TVNewsProgram a owl:Class ; rdfs:subClassOf px:InformationContent .

px:OpinionArticle a owl:Class ; rdfs:subClassOf px:InformationContent .

px:AudienceSegment a owl:Class .

px:DemographicGroup a owl:Class ; rdfs:subClassOf px:AudienceSegment .

px:AgeCohort a owl:Class ; rdfs:subClassOf px:DemographicGroup .

px:SocioEconomicGroup a owl:Class ; rdfs:subClassOf px:DemographicGroup .

px:ViewingBehaviorProfile a owl:Class ; rdfs:subClassOf px:AudienceSegment
.

px:BingeWatcherProfile a owl:Class ; rdfs:subClassOf
px:ViewingBehaviorProfile .

px:CasualViewerProfile a owl:Class ; rdfs:subClassOf
px:ViewingBehaviorProfile .

px:Impact a owl:Class .

px:CulturalImpact a owl:Class ; rdfs:subClassOf px:Impact .

px:CulturalHomogenizationImpact a owl:Class ; rdfs:subClassOf
px:CulturalImpact .

```

```

px:CulturalDiversificationImpact a owl:Class ; rdfs:subClassOf
px:CulturalImpact .

px:BehavioralImpact a owl:Class ; rdfs:subClassOf px:Impact .

px:ViewingHabitChangeImpact a owl:Class ; rdfs:subClassOf
px:BehavioralImpact .

px:PoliticalEngagementImpact a owl:Class ; rdfs:subClassOf
px:BehavioralImpact .

px:GeopoliticalImpact a owl:Class ; rdfs:subClassOf px:Impact .

px:SoftPowerImpact a owl:Class ; rdfs:subClassOf px:GeopoliticalImpact .

px:RegulatoryTensionImpact a owl:Class ; rdfs:subClassOf
px:GeopoliticalImpact .

px:GovernanceAndPolicy a owl:Class .

px:MediaRegulationPolicy a owl:Class ; rdfs:subClassOf
px:GovernanceAndPolicy .

px:StreamingRegulationPolicy a owl:Class ; rdfs:subClassOf
px:MediaRegulationPolicy .

px:BroadcastingRegulationPolicy a owl:Class ; rdfs:subClassOf
px:MediaRegulationPolicy .

px:PlatformGovernanceModel a owl:Class ; rdfs:subClassOf
px:GovernanceAndPolicy .

px:AlgorithmTransparencyPolicy a owl:Class ; rdfs:subClassOf
px:PlatformGovernanceModel .

px:TechnologicalFeature a owl:Class .

px:RecommendationAlgorithm a owl:Class ; rdfs:subClassOf
px:TechnologicalFeature .

px:CollaborativeFilteringAlgorithm a owl:Class ; rdfs:subClassOf
px:RecommendationAlgorithm .

px:ContentPromotionAlgorithm a owl:Class ; rdfs:subClassOf
px:RecommendationAlgorithm .

px:Country a owl:Class .

px:OriginCountry a owl:Class ; rdfs:subClassOf px:Country .

px:ContentOriginCountry a owl:Class ; rdfs:subClassOf px:OriginCountry .

px:MarketCountry a owl:Class ; rdfs:subClassOf px:Country .

px:TargetMarketCountry a owl:Class ; rdfs:subClassOf px:MarketCountry .

```

--- Object Properties ---

px:platformDistributesContent a owl:ObjectProperty ; rdfs:domain
px:StreamingPlatform ; rdfs:range px:MediaContent .

px:traditionalOutletBroadcastsContent a owl:ObjectProperty ; rdfs:domain
px:TraditionalMediaOutlet ; rdfs:range px:MediaContent .

px:contentProducedByPlatform a owl:ObjectProperty ; rdfs:domain
px:MediaContent ; rdfs:range px:MediaPlatform .

px:contentTargetsAudienceSegment a owl:ObjectProperty ; rdfs:domain
px:MediaContent ; rdfs:range px:AudienceSegment .

px:audienceConsumesContent a owl:ObjectProperty ; rdfs:domain
px:AudienceSegment ; rdfs:range px:MediaContent .

px:audiencePrefersPlatform a owl:ObjectProperty ; rdfs:domain
px:AudienceSegment ; rdfs:range px:MediaPlatform .

px:impactAffectsAudienceSegment a owl:ObjectProperty ; rdfs:domain
px:Impact ; rdfs:range px:AudienceSegment .

px:impactObservedInCountry a owl:ObjectProperty ; rdfs:domain px:Impact ;
rdfs:range px:Country .

px:contentSetInCountry a owl:ObjectProperty ; rdfs:domain px:MediaContent ;
rdfs:range px:Country .

px:platformOperatesInMarketCountry a owl:ObjectProperty ; rdfs:domain
px:StreamingPlatform ; rdfs:range px:MarketCountry .

px:contentOriginatesFromCountry a owl:ObjectProperty ; rdfs:domain
px:MediaContent ; rdfs:range px:OriginCountry .

px:regulationAppliesToPlatform a owl:ObjectProperty ; rdfs:domain
px:MediaRegulationPolicy ; rdfs:range px:MediaPlatform .

px:regulationOriginCountry a owl:ObjectProperty ; rdfs:domain
px:MediaRegulationPolicy ; rdfs:range px:OriginCountry .

px:policyRespondsToImpact a owl:ObjectProperty ; rdfs:domain
px:MediaRegulationPolicy ; rdfs:range px:Impact .

px:platformImplementsAlgorithm a owl:ObjectProperty ; rdfs:domain
px:StreamingPlatform ; rdfs:range px:RecommendationAlgorithm .

px:algorithmShapesBehavioralImpact a owl:ObjectProperty ; rdfs:domain
px:RecommendationAlgorithm ; rdfs:range px:BehavioralImpact .

px:algorithmPersonalizesForAudience a owl:ObjectProperty ; rdfs:domain
px:RecommendationAlgorithm ; rdfs:range px:AudienceSegment .

px:softPowerDirectedAtCountry a owl:ObjectProperty ; rdfs:domain
px:SoftPowerImpact ; rdfs:range px:MarketCountry .

```
px:traditionalOutletCompetesWithPlatform a owl:ObjectProperty ; rdfs:domain
px:TraditionalMediaOutlet ; rdfs:range px:StreamingPlatform .
```

```
px:platformGovernedByModel a owl:ObjectProperty ; rdfs:domain
px:MediaPlatform ; rdfs:range px:PlatformGovernanceModel .
```

```
# --- Data Properties ---
```

```
px:platformName a owl:DatatypeProperty ; rdfs:domain px:MediaPlatform ;
rdfs:range xsd:string .
```

```
px:launchYear a owl:DatatypeProperty ; rdfs:domain px:MediaPlatform ;
rdfs:range xsd:integer .
```

```
px:headquartersLocation a owl:DatatypeProperty ; rdfs:domain
px:MediaPlatform ; rdfs:range xsd:string .
```

```
px:subscriberCount a owl:DatatypeProperty ; rdfs:domain
px:StreamingPlatform ; rdfs:range xsd:integer .
```

```
px:averageWatchTimePerDay a owl:DatatypeProperty ; rdfs:domain
px:StreamingPlatform ; rdfs:range xsd:float .
```

```
px:isGlobalService a owl:DatatypeProperty ; rdfs:domain
px:StreamingPlatform ; rdfs:range xsd:boolean .
```

```
px:channelName a owl:DatatypeProperty ; rdfs:domain
px:BroadcastTelevisionChannel ; rdfs:range xsd:string .
```

```
px:circulationVolume a owl:DatatypeProperty ; rdfs:domain px:PrintNewspaper
; rdfs:range xsd:integer .
```

```
px:radioFrequencyMHz a owl:DatatypeProperty ; rdfs:domain px:RadioStation ;
rdfs:range xsd:float .
```

```
px:contentTitle a owl:DatatypeProperty ; rdfs:domain px:MediaContent ;
rdfs:range xsd:string .
```

```
px:releaseDate a owl:DatatypeProperty ; rdfs:domain px:MediaContent ;
rdfs:range xsd:date .
```

```
px:genreLabel a owl:DatatypeProperty ; rdfs:domain px:EntertainmentContent
; rdfs:range xsd:string .
```

```
px:isOriginalProduction a owl:DatatypeProperty ; rdfs:domain
px:StreamingOriginalSeries ; rdfs:range xsd:boolean .
```

```
px:ageGroupLabel a owl:DatatypeProperty ; rdfs:domain px:AgeCohort ;
rdfs:range xsd:string .
```

```
px:incomeBracket a owl:DatatypeProperty ; rdfs:domain px:SocioEconomicGroup
; rdfs:range xsd:string .
```

```
px:bingeSessionLengthMinutes a owl:DatatypeProperty ; rdfs:domain
px:BingeWatcherProfile ; rdfs:range xsd:integer .
```

```
px:policyName a owl:DatatypeProperty ; rdfs:domain px:MediaRegulationPolicy ; rdfs:range xsd:string .
```

```
px:enforcementStartDate a owl:DatatypeProperty ; rdfs:domain px:MediaRegulationPolicy ; rdfs:range xsd:date .
```

```
px:impactIntensityScore a owl:DatatypeProperty ; rdfs:domain px:Impact ; rdfs:range xsd:float .
```

```
px:cosmopolitanismIndex a owl:DatatypeProperty ; rdfs:domain px:AudienceSegment ; rdfs:range xsd:float .
```

```
# --- Instances & Revised Triples ---
```

```
px:NetflixPlatform a px:SubscriptionStreamingPlatform .
```

```
px:DisneyPlusPlatform a px:SubscriptionStreamingPlatform .
```

```
px:YouTubePlatform a px:AdSupportedStreamingPlatform .
```

```
px:Kan11Channel a px:BroadcastTelevisionChannel .
```

```
px:HaaretzPrintEdition a px:PrintNewspaper .
```

```
px:BBCOneChannel a px:BroadcastTelevisionChannel .
```

```
px:SquidGameSeriesContent a px:StreamingOriginalSeries .
```

```
px:FaudaSeriesContent a px:StreamingOriginalSeries .
```

```
px:EveningNewsBBCProgram a px:TVNewsProgram .
```

```
px:HaaretzOpinionArticleOnline a px:OpinionArticle .
```

```
px:IsraeliYoungAdults18to34Group a px:AgeCohort .
```

```
px:GlobalMiddleClassViewersGroup a px:SocioEconomicGroup .
```

```
px:HeavyBingeWatchersClusterProfile a px:BingeWatcherProfile .
```

```
px:NetflixCulturalHomogenizationGlobalImpact a  
px:CulturalHomogenizationImpact .
```

```
px:NetflixCulturalDiversificationIsraelImpact a  
px:CulturalDiversificationImpact .
```

```
px:StreamingSoftPowerUSImpact a px:SoftPowerImpact .
```

```
px:EUAVMSDStreamingPolicy a px:StreamingRegulationPolicy .
```

```
px:IsraelBroadcastingRegulationPolicy a px:BroadcastingRegulationPolicy .
```

```
px:NetflixMainRecommendationAlgorithm a px:CollaborativeFilteringAlgorithm .
```

px:UnitedStatesOriginCountryInstance a px:ContentOriginCountry .

px:NetflixPlatform px:platformDistributesContent px:SquidGameSeriesContent .

px:YouTubePlatform px:traditionalOutletBroadcastsContent
px:HaaretzOpinionArticleOnline .

px:BBCOneChannel px:contentProducedByPlatform px:EveningNewsBBCProgram .

px:HaaretzPrintEdition px:contentTargetsAudienceSegment
px:IsraeliYoungAdults18to34Group .

px:HeavyBingeWatchersClusterProfile px:audienceConsumesContent
px:SquidGameSeriesContent .

px:GlobalMiddleClassViewersGroup px:audiencePrefersPlatform
px:NetflixPlatform .

px:NetflixCulturalHomogenizationGlobalImpact
px:impactAffectsAudienceSegment px:GlobalMiddleClassViewersGroup .

px:NetflixCulturalDiversificationIsraelImpact px:impactObservedInCountry
px:UnitedStatesOriginCountryInstance .

px:FaudaSeriesContent px:contentSetInCountry
px:UnitedStatesOriginCountryInstance .

px:NetflixPlatform px:platformOperatesInMarketCountry
px:UnitedStatesOriginCountryInstance .

px:SquidGameSeriesContent px:contentOriginatesFromCountry
px:UnitedStatesOriginCountryInstance .

px:EUAVMSDStreamingPolicy px:regulationAppliesToPlatform px:NetflixPlatform .

px:IsraelBroadcastingRegulationPolicy px:regulationOriginCountry
px:UnitedStatesOriginCountryInstance .

px:EUAVMSDStreamingPolicy px:policyRespondsToImpact
px:NetflixCulturalHomogenizationGlobalImpact .

px:NetflixPlatform px:platformImplementsAlgorithm
px:NetflixMainRecommendationAlgorithm .

px:NetflixMainRecommendationAlgorithm px:algorithmShapesBehavioralImpact
px:ViewingHabitChangeImpact .

px:NetflixMainRecommendationAlgorithm px:algorithmPersonalizesForAudience
px:IsraeliYoungAdults18to34Group .

px:StreamingSoftPowerUSImpact px:softPowerDirectedAtCountry
px:UnitedStatesOriginCountryInstance .

```
px:Kan11Channel px:traditionalOutletCompetesWithPlatform px:NetflixPlatform
.
```

```
px:NetflixPlatform px:platformGovernedByModel
px:AlgorithmTransparencyPolicy .
```

Appendix C: Model 3 - ChatGPT RDF/OWL Code (ChatGPT_KG.ttl)

```
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
```

```
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
```

```
@prefix owl: <http://www.w3.org/2002/07/owl#> .
```

```
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
```

```
@prefix gpt: <http://example.org/chatgpt#> .
```

```
<http://example.org/chatgpt> a owl:Ontology ;
```

```
    rdfs:label "ChatGPT Comprehensive Media & Soft Power Ontology" .
```

```
# --- Top Level Classes ---
```

```
gpt:MediaEntity a owl:Class .
```

```
gpt:MediaPlatform a owl:Class ; rdfs:subClassOf gpt:MediaEntity .
```

```
gpt:GlobalStreamingPlatform a owl:Class ; rdfs:subClassOf gpt:MediaPlatform
.
```

```
gpt:RegionalStreamingPlatform a owl:Class ; rdfs:subClassOf
gpt:MediaPlatform .
```

```
gpt:SocialVideoPlatform a owl:Class ; rdfs:subClassOf gpt:MediaPlatform .
```

```
gpt:TraditionalMediaOrganization a owl:Class ; rdfs:subClassOf
gpt:MediaEntity .
```

```
gpt:BroadcastTelevisionNetwork a owl:Class ; rdfs:subClassOf
gpt:TraditionalMediaOrganization .
```

```
gpt:CableTelevisionNetwork a owl:Class ; rdfs:subClassOf
gpt:TraditionalMediaOrganization .
```

```
gpt:PublicServiceBroadcaster a owl:Class ; rdfs:subClassOf
gpt:TraditionalMediaOrganization .
```

```
gpt:NewspaperOrganization a owl:Class ; rdfs:subClassOf
gpt:TraditionalMediaOrganization .
```

```
gpt:MediaContent a owl:Class .
```

```
gpt:AudiovisualContent a owl:Class ; rdfs:subClassOf gpt:MediaContent .
```

`gpt:Film a owl:Class ; rdfs:subClassOf gpt:AudiovisualContent .`
`gpt:TelevisionSeries a owl:Class ; rdfs:subClassOf gpt:AudiovisualContent .`
`gpt:NewsProgram a owl:Class ; rdfs:subClassOf gpt:AudiovisualContent .`
`gpt:Impact a owl:Class .`
`gpt:CulturalImpact a owl:Class ; rdfs:subClassOf gpt:Impact .`
`gpt:CulturalHomogenization a owl:Class ; rdfs:subClassOf gpt:CulturalImpact .`
`gpt:CulturalHybridization a owl:Class ; rdfs:subClassOf gpt:CulturalImpact .`
`gpt:GeopoliticalImpact a owl:Class ; rdfs:subClassOf gpt:Impact .`
`gpt:SoftPowerImpact a owl:Class ; rdfs:subClassOf gpt:GeopoliticalImpact .`
`gpt:BehavioralImpact a owl:Class ; rdfs:subClassOf gpt:Impact .`
`gpt:BingeWatchingBehavior a owl:Class ; rdfs:subClassOf gpt:BehavioralImpact .`
`gpt:AudienceFragmentation a owl:Class ; rdfs:subClassOf gpt:BehavioralImpact .`
`gpt:EconomicImpact a owl:Class ; rdfs:subClassOf gpt:Impact .`
`gpt:LocalProductionImpact a owl:Class ; rdfs:subClassOf gpt:EconomicImpact .`
`gpt:Actor a owl:Class .`
`gpt:InstitutionalActor a owl:Class ; rdfs:subClassOf gpt:Actor .`
`gpt:RegulatoryAuthority a owl:Class ; rdfs:subClassOf gpt:InstitutionalActor .`
`gpt:AudienceCommunity a owl:Class ; rdfs:subClassOf gpt:Actor .`
`gpt:CulturalCommunity a owl:Class ; rdfs:subClassOf gpt:Actor .`
`gpt:GeographicEntity a owl:Class .`
`gpt:Country a owl:Class ; rdfs:subClassOf gpt:GeographicEntity .`
`gpt:DemocraticCountry a owl:Class ; rdfs:subClassOf gpt:Country .`
`gpt:Policy a owl:Class .`
`gpt:MediaPolicy a owl:Class ; rdfs:subClassOf gpt:Policy .`

```

gpt:ContentQuotaPolicy a owl:Class ; rdfs:subClassOf gpt:MediaPolicy .

gpt:Technology a owl:Class .

gpt:AlgorithmicTechnology a owl:Class ; rdfs:subClassOf gpt:Technology .

gpt:RecommendationAlgorithm a owl:Class ; rdfs:subClassOf
gpt:AlgorithmicTechnology .

gpt:AudienceMeasurementTechnology a owl:Class ; rdfs:subClassOf
gpt:Technology .

gpt:TelevisionRatingSystem a owl:Class ; rdfs:subClassOf
gpt:AudienceMeasurementTechnology .

# --- Object Properties ---

gpt:distributesContent a owl:ObjectProperty ; rdfs:domain gpt:MediaPlatform
; rdfs:range gpt:MediaContent .

gpt:ownsPlatform a owl:ObjectProperty ; rdfs:domain gpt:InstitutionalActor
; rdfs:range gpt:MediaPlatform .

gpt:operatesInCountry a owl:ObjectProperty ; rdfs:domain gpt:MediaPlatform
; rdfs:range gpt:Country .

gpt:targetsAudience a owl:ObjectProperty ; rdfs:domain gpt:MediaContent ;
rdfs:range gpt:AudienceCommunity .

gpt:usesTechnology a owl:ObjectProperty ; rdfs:domain gpt:MediaPlatform ;
rdfs:range gpt:Technology .

gpt:recommendsContent a owl:ObjectProperty ; rdfs:domain
gpt:RecommendationAlgorithm ; rdfs:range gpt:MediaContent .

gpt:causesImpact a owl:ObjectProperty ; rdfs:domain gpt:MediaEntity ;
rdfs:range gpt:Impact .

gpt:contentCausesImpact a owl:ObjectProperty ; rdfs:domain gpt:MediaContent
; rdfs:range gpt:Impact .

gpt:influencesViewingBehavior a owl:ObjectProperty ; rdfs:domain
gpt:MediaPlatform ; rdfs:range gpt:BehavioralImpact .

gpt:servesAsSoftPowerFor a owl:ObjectProperty ; rdfs:domain
gpt:MediaContent ; rdfs:range gpt:Country .

gpt:exportsCulturalValuesTo a owl:ObjectProperty ; rdfs:domain
gpt:MediaContent ; rdfs:range gpt:Country .

gpt:representsCommunity a owl:ObjectProperty ; rdfs:domain gpt:MediaContent
; rdfs:range gpt:CulturalCommunity .

gpt:regulatedBy a owl:ObjectProperty ; rdfs:domain gpt:MediaPlatform ;
rdfs:range gpt:RegulatoryAuthority .

```

```
gpt:appliesToPlatform a owl:ObjectProperty ; rdfs:domain gpt:Policy ;  
rdfs:range gpt:MediaPlatform .
```

```
gpt:requiresLocalContent a owl:ObjectProperty ; rdfs:domain  
gpt:ContentQuotaPolicy ; rdfs:range gpt:MediaPlatform .
```

```
gpt:competesWith a owl:ObjectProperty ; rdfs:domain gpt:MediaEntity ;  
rdfs:range gpt:MediaEntity .
```

```
gpt:shapesPublicOpinionIn a owl:ObjectProperty ; rdfs:domain  
gpt:MediaEntity ; rdfs:range gpt:Country .
```

```
gpt:measuredBy a owl:ObjectProperty ; rdfs:domain gpt:MediaContent ;  
rdfs:range gpt:AudienceMeasurementTechnology .
```

```
# --- Selected Instances & Triples ---
```

```
gpt:Netflix a gpt:GlobalStreamingPlatform .
```

```
gpt:DisneyPlus a gpt:GlobalStreamingPlatform .
```

```
gpt:YouTube a gpt:SocialVideoPlatform .
```

```
gpt:BBCOne a gpt:PublicServiceBroadcaster .
```

```
gpt:CNN a gpt:CableTelevisionNetwork .
```

```
gpt:Kan11 a gpt:PublicServiceBroadcaster .
```

```
gpt:SquidGame a gpt:TelevisionSeries .
```

```
gpt:StrangerThings a gpt:TelevisionSeries .
```

```
gpt:MoneyHeist a gpt:TelevisionSeries .
```

```
gpt:TheCrown a gpt:TelevisionSeries .
```

```
gpt:Fauda a gpt:TelevisionSeries .
```

```
gpt:SouthKorea a gpt:DemocraticCountry .
```

```
gpt:UnitedStates a gpt:DemocraticCountry .
```

```
gpt:UnitedKingdom a gpt:DemocraticCountry .
```

```
gpt:Israel a gpt:DemocraticCountry .
```

```
gpt:EuropeanUnionMediaRegulator a gpt:RegulatoryAuthority .
```

```
gpt:EuropeanContentQuota a gpt:ContentQuotaPolicy .
```

```
gpt:NetflixRecommendationEngine a gpt:RecommendationAlgorithm .
```

```
gpt:NielsenRatingSystem a gpt:TelevisionRatingSystem .
```

gpt:KoreanWaveCommunity a gpt:CulturalCommunity .
 gpt:GlobalBingeWatchingGrowth a gpt:BingeWatchingBehavior .
 gpt:KoreanSoftPowerGrowth a gpt:SoftPowerImpact .
 gpt:LocalTelevisionAudienceFragmentation a gpt:AudienceFragmentation .
 gpt:NetflixInc a gpt:InstitutionalActor .
 gpt:NetflixInc gpt:ownsPlatform gpt:Netflix .
 gpt:Netflix gpt:operatesInCountry gpt:UnitedStates .
 gpt:Netflix gpt:operatesInCountry gpt:Israel .
 gpt:Netflix gpt:distributesContent gpt:SquidGame .
 gpt:Netflix gpt:distributesContent gpt:StrangerThings .
 gpt:Netflix gpt:distributesContent gpt:Fauda .
 gpt:SquidGame gpt:servesAsSoftPowerFor gpt:SouthKorea .
 gpt:Fauda gpt:exportsCulturalValuesTo gpt:UnitedStates .
 gpt:SquidGame gpt:representsCommunity gpt:KoreanWaveCommunity .
 gpt:SquidGame gpt:contentCausesImpact gpt:KoreanSoftPowerGrowth .
 gpt:Netflix gpt:influencesViewingBehavior gpt:GlobalBingeWatchingGrowth .
 gpt:Netflix gpt:causesImpact gpt:LocalTelevisionAudienceFragmentation .
 gpt:Netflix gpt:usesTechnology gpt:NetflixRecommendationEngine .
 gpt:NetflixRecommendationEngine gpt:recommendsContent gpt:SquidGame .
 gpt:Netflix gpt:regulatedBy gpt:EuropeanUnionMediaRegulator .
 gpt:EuropeanContentQuota gpt:appliesToPlatform gpt:Netflix .
 gpt:EuropeanContentQuota gpt:requiresLocalContent gpt:Netflix .
 gpt:Netflix gpt:competesWith gpt:DisneyPlus .
 gpt:Netflix gpt:competesWith gpt:BBCOne .
 gpt:Kan11 gpt:shapesPublicOpinionIn gpt:Israel .

Appendix D: Neo4j Cypher Queries & Database Results

Competency Question 1: Content-Driven Conflicts & Competition

General Research Goal: Identify streaming platforms and content items that trigger geopolitical conflicts, diplomatic tensions, or direct industry competition.

Model 1 (Gemini Cypher):

```
MATCH (platform)-[:producesContent]->(content)-[:triggersConflict]->(conflict)
```

```
RETURN platform.name AS Platform, content.title AS Content, conflict.name AS Conflict
```



The screenshot shows a Neo4j Cypher query execution interface. The query is displayed in a text area with line numbers 1 and 2. Below the query, there are buttons for 'Table' and 'RAW'. To the right of the buttons are search, dropdown, and download icons. Below the buttons, a table displays the results of the query. The table has three columns: 'Platform', 'Content', and 'Conflict'. The first row of data shows 'Netflix' for Platform, 'Fauda' for Content, and 'Diplomatic_Row_Fauda' for Conflict.

```
1 MATCH (platform)-[:producesContent]->(content)-  
  [:triggersConflict]->(conflict)  
2 RETURN platform.name AS Platform, content.title AS  
  Content, conflict.name AS Conflict
```

Platform	Content	Conflict
"Netflix"	"Fauda"	"Diplomatic_Row_Fauda"

Figure D.1: Gemini Cypher query execution and deterministic conflict path result.

Model 2 (Perplexity Cypher):

```
// Find streaming platforms and content tied to geopolitical/diplomatic impacts or  
competition
```

```
MATCH (sp:StreamingPlatform)
```

```
OPTIONAL MATCH (sp)-[:DISTRIBUTES_CONTENT]->(content:MediaContent)
```

```
OPTIONAL MATCH (sp)<-[:APPLIES_TO_PLATFORM]-(policy:GovernanceAndPolicy)
```

```
OPTIONAL MATCH (policy)-[:RESPONDS_TO_IMPACT]->(impact:Impact)
```

```
OPTIONAL MATCH (impact)-[:OBSERVED_IN_COUNTRY]->(country:Country)
```

```
OPTIONAL MATCH (sp)<-[:COMPETES_WITH_PLATFORM]-(tmo:TraditionalMediaOutlet)
```

```
OPTIONAL MATCH (sp)-[:IMPLEMENTS_ALGORITHM]->(alg:TechnologicalFeature)
```

```
OPTIONAL MATCH (alg)-[:SHAPES_BEHAVIORAL_IMPACT]->(behavioral:Impact)
```

```
// Focus on geopolitical/diplomatic signals:
```

```

// - policies responding to impacts, or

// - explicit competition from traditional outlets

WHERE impact IS NOT NULL OR tmo IS NOT NULL

RETURN

  sp.id AS platformId,

  sp.name AS streamingPlatform,

  collect(DISTINCT content.title) AS relatedContent,

  collect(DISTINCT policy.name) AS relatedPolicies,

  collect(DISTINCT {impactId: impact.id, impactType: impact.type}) AS
impacts,

  collect(DISTINCT country.name) AS affectedCountries,

  collect(DISTINCT tmo.name) AS competingTraditionalOutlets

;

```

```

1 // Find streaming platforms and content tied to geopolitical/diplomatic impacts or competition
2 MATCH (sp:StreamingPlatform)
3 OPTIONAL MATCH (sp)-[:DISTRIBUTES_CONTENT]->(content:MediaContent)
4 OPTIONAL MATCH (sp)-[:APPLIES_TO_PLATFORM]-(policy:GovernanceAndPolicy)
5 OPTIONAL MATCH (policy)-[:RESPONDS_TO_IMPACT]->(impact:Impact)
6 OPTIONAL MATCH (impact)-[:OBSERVED_IN_COUNTRY]->(country:Country)
7 OPTIONAL MATCH (sp)-[:COMPETES_WITH_PLATFORM]-(tmo:TraditionalMediaOutlet)
8 OPTIONAL MATCH (sp)-[:IMPLEMENTS_ALGORITHM]->(alg:TechnologicalFeature)
9 OPTIONAL MATCH (alg)-[:SHAPES_BEHAVIORAL_IMPACT]->(behavioral:Impact)

10
11 // Focus on geopolitical/diplomatic signals:
12 // - policies responding to impacts, or
13 // - explicit competition from traditional outlets
14 WHERE impact IS NOT NULL OR tmo IS NOT NULL
15
16 RETURN
17   sp.id AS platformId,
18   sp.name AS streamingPlatform,
19   collect(DISTINCT content.title) AS relatedContent,
20   collect(DISTINCT policy.name) AS relatedPolicies,
21   collect(DISTINCT {impactId: impact.id, impactType: impact.type}) AS impacts,
22   collect(DISTINCT country.name) AS affectedCountries,
23   collect(DISTINCT tmo.name) AS competingTraditionalOutlets
24 ;

```

platformId	streamingPlatform	relatedContent	relatedPolicies	impacts	affectedCountries	competingTraditionalOutlets
"NetflixPlatform"	"Netflix"	["Squid Game"]	["EU AVMSD Policy"]	[{ impactId:"NetflixCul turalHomogenizationG lobalImpact", impactType:"Homogeni zation" }]	[]	["Kan 11"]

Figure D.2: Perplexity Cypher aggregation query and multi-dimensional output.

Model 3 (ChatGPT Cypher):

```
MATCH (platform)-[:ns0__competesWith]->(competitor)
RETURN platform.uri AS Platform,
       competitor.uri AS Competitor;
```



The screenshot shows the Neo4j Cypher query interface. The query entered is `neo4j$ MATCH ()-[r]→() RETURN DISTINCT type(r) ORDER BY type(r);`. The results are displayed in a table with one column, `type(r)`, listing seven relationship types: `"ns0__appliesToPlatform"`, `"ns0__causesImpact"`, `"ns0__competesWith"`, `"ns0__contentCausesImpact"`, `"ns0__distributesContent"`, `"ns0__exportsCulturalValuesTo"`, and `"ns0__hasParent"`. A status bar at the bottom indicates "Started streaming 19 records after 106 ms and completed after 112 ms."

type(r)
"ns0__appliesToPlatform"
"ns0__causesImpact"
"ns0__competesWith"
"ns0__contentCausesImpact"
"ns0__distributesContent"
"ns0__exportsCulturalValuesTo"
"ns0__hasParent"



The screenshot shows the Neo4j Cypher query interface. The query entered is `neo4j$ MATCH (platform)-[:ns0__competesWith]→(competitor) RETURN platform.uri AS Platform, competitor.uri AS Competitor;`. The results are displayed in a table with two columns: `Platform` and `Competitor`. There are two rows of results. A status bar at the bottom indicates "Started streaming 2 records after 21 ms and completed after 25 ms."

Platform	Competitor
"http://example.org/chatgpt#Netflix"	"http://example.org/chatgpt#DisneyPlus"
"http://example.org/chatgpt#Netflix"	"http://example.org/chatgpt#BBCOne"

Figure D.3: ChatGPT schema property check and competition query output in Neo4j

Competency Question 2: Geopolitical Regulation & Content Restrictions

General Research Goal: Identify geographic regions, governing authorities, and regulatory policies enforcing censorship or quotas on streaming services.

Model 1 (Gemini Cypher):

```
MATCH (region)-[:enforcesPolicy]->(censorship)-[:restrictsContent]->(content)<-[:producesContent]-(platform)
RETURN region.name AS Region,
       censorship.name AS CensorshipPolicy,
```

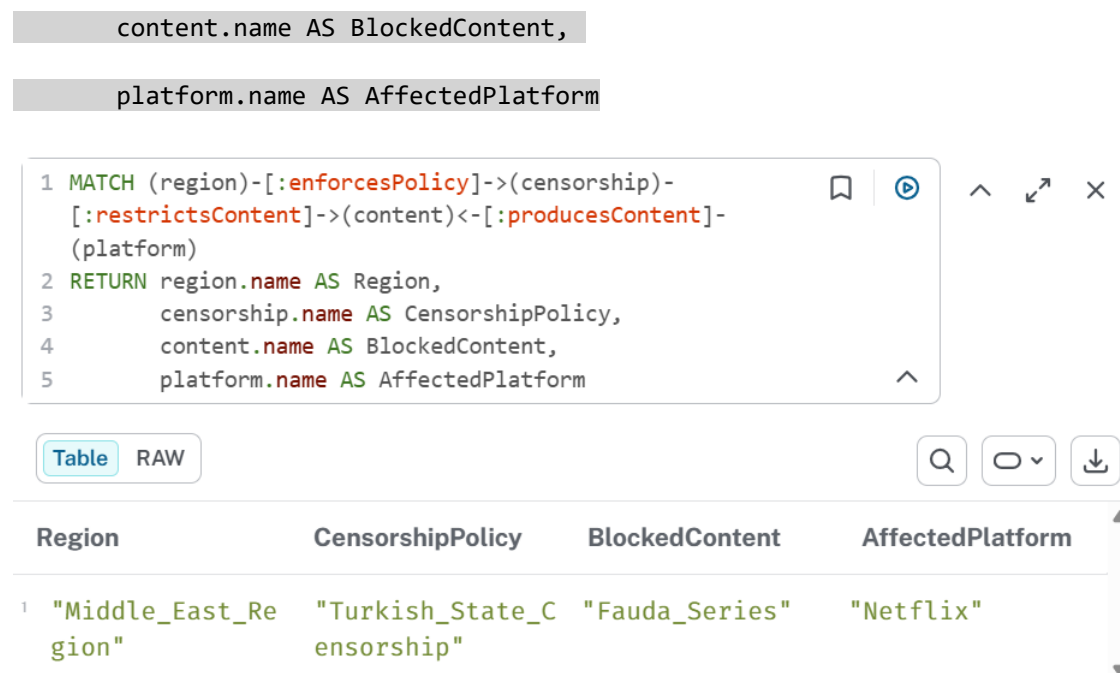


Figure D.4: Gemini four-node traversal identifying regional censorship policies and restricted content.

Model 2 (Perplexity Cypher):

```

// Find countries, policies, and streaming platforms subject to media
regulation (e.g., censorship/quotas)

MATCH (policy:GovernanceAndPolicy)-[:APPLIES_TO_PLATFORM]-
>(sp:StreamingPlatform)

MATCH (policy)-[:REGULATION_ORIGIN_COUNTRY]->(country:Country)

OPTIONAL MATCH (policy)-[:RESPONDS_TO_IMPACT]->(impact:Impact)

RETURN

country.name AS governingCountry, policy.id AS policyId, policy.name AS
policyName,

collect(DISTINCT sp.name) AS affectedStreamingPlatforms,

collect(DISTINCT {impactId: impact.id, impactType: impact.type}) AS
policyTriggers;

```

```

1 // Find countries, policies, and streaming platforms subject to media regulation (e.g., censorship/quotas)
2 MATCH (policy:GovernanceAndPolicy)-[:APPLIES_TO_PLATFORM]->(sp:StreamingPlatform)
3 MATCH (policy)-[:REGULATION_ORIGIN_COUNTRY]->(country:Country)
4 OPTIONAL MATCH (policy)-[:RESPONDS_TO_IMPACT]->(impact:Impact)
5
6 RETURN
7   country.name AS governingCountry, policy.id AS policyId, policy.name AS policyName,
8   collect(DISTINCT sp.name) AS affectedStreamingPlatforms,
9   collect(DISTINCT {impactId: impact.id, impactType: impact.type}) AS policyTriggers;

```

Figure D.5: Perplexity regulatory query execution returning zero records due to disjoint schema paths.

Model 3 (ChatGPT Cypher): •

```

MATCH (policy)-[:ns0__appliesToPlatform]->(platform)-
[:ns0__regulatedBy]->(regulator)

RETURN policy.uri AS Policy,
        platform.uri AS Platform,
        regulator.uri AS Regulator;

```

The screenshot shows the Neo4j Cypher query editor with the following query:

```
neo4j$ MATCH (policy)-[:ns0__appliesToPlatform]->(platform)-[:ns0__regulatedBy]->(regulator) RETURN ...
```

The results are displayed in a table with three columns: Policy, Platform, and Regulator. The first row contains the following URIs:

Policy	Platform	Regulator
"http://example.org/chatgpt#EuropeanContentQuota"	"http://example.org/chatgpt#Netflix"	"http://example.org/chatgpt#EuropeanUnionMediaRegulator"

At the bottom, a status message reads: "Started streaming 1 records after 22 ms and completed after 26 ms."

Figure D.6: ChatGPT tripartite regulatory relationship execution in Neo4j.

Competency Question 3: Behavioral Shifts & Demographic Audience Impact

General Research Goal: Map the demographic groups or audience segments experiencing socio-behavioral shifts (such as binge-watching, sleep disruption, or audience fragmentation) caused by platforms.

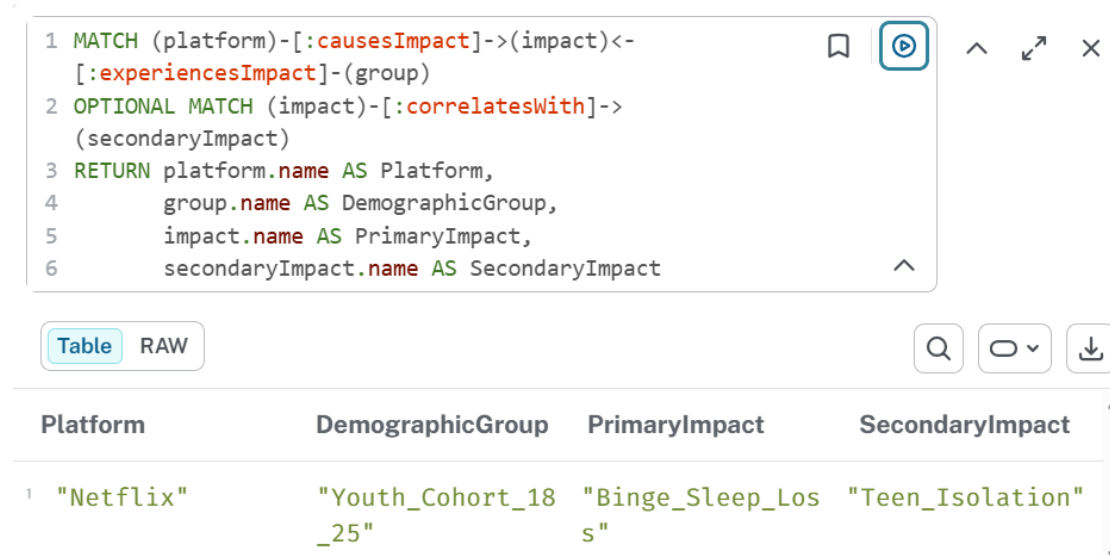
Model 1 (Gemini Cypher):

```

MATCH (platform)-[:causesImpact]->(impact)<-[:experiencesImpact]-(group)
OPTIONAL MATCH (impact)-[:correlatesWith]->(secondaryImpact)

RETURN platform.name AS Platform,
        group.name AS DemographicGroup,
        impact.name AS PrimaryImpact,
        secondaryImpact.name AS SecondaryImpact

```



```

1 MATCH (platform)-[:causesImpact]->(impact)<-
  [:experiencesImpact]-(group)
2 OPTIONAL MATCH (impact)-[:correlatesWith]->
  (secondaryImpact)
3 RETURN platform.name AS Platform,
4         group.name AS DemographicGroup,
5         impact.name AS PrimaryImpact,
6         secondaryImpact.name AS SecondaryImpact

```

Platform	DemographicGroup	PrimaryImpact	SecondaryImpact
"Netflix"	"Youth_Cohort_18 _25"	"Binge_Sleep_Los s"	"Teen_Isolation"

Figure D.7: Gemini behavioral impact and secondary correlation path output.

Model 2 (Perplexity Cypher):

```

// Find audience segments experiencing behavioral impacts driven by
platforms/algorithms

MATCH (sp:StreamingPlatform)-[:IMPLEMENTS_ALGORITHM]-
>(alg:TechnologicalFeature)

MATCH (alg)-[:SHAPES_BEHAVIORAL_IMPACT]->(behavioral:Impact)

MATCH (alg)-[:PERSONALIZES_FOR_AUDIENCE]->(aud:AudienceSegment)

OPTIONAL MATCH (aud)-[:CONSUMES_CONTENT]->(content:MediaContent)

OPTIONAL MATCH (aud)-[:PREFERS_PLATFORM]->(prefPlatform:MediaPlatform)

// Optional focus on binge-watching or habit-change impacts:

// WHERE aud:BingeWatcherProfile OR behavioral.id =
'ViewingHabitChangeImpact'

RETURN
sp.name AS platform,
alg.id AS algorithmId,
behavioral.id AS behavioralImpactId,
aud.id AS audienceSegmentId,
aud.label AS audienceLabel,
aud.bingeSessionLengthMinutes AS bingeSessionLengthMinutes,
collect(DISTINCT content.title) AS consumedContent,
prefPlatform.name AS preferredPlatform;

```

```

1 // Find audience segments experiencing behavioral impacts driven by platforms/algorithms
2 MATCH (sp:StreamingPlatform)-[:IMPLEMENTS_ALGORITHM]->(alg:TechnologicalFeature)
3 MATCH (alg)-[:SHAPES_BEHAVIORAL_IMPACT]->(behavioral:Impact)
4 MATCH (alg)-[:PERSONALIZES_FOR_AUDIENCE]->(aud:AudienceSegment)
5 OPTIONAL MATCH (aud)-[:CONSUMES_CONTENT]->(content:MediaContent)
6 OPTIONAL MATCH (aud)-[:PREFERS_PLATFORM]->(prefPlatform:MediaPlatform)
7 |
8 // Optional focus on binge-watching or habit-change impacts:
9 // WHERE aud:BingeWatcherProfile OR behavioral.id = 'ViewingHabitChangeImpact'

10 RETURN
11   sp.name AS platform,
12   alg.id AS algorithmId,
13   behavioral.id AS behavioralImpactId,
14   aud.id AS audienceSegmentId,
15   aud.label AS audienceLabel,
16   aud.bingeSessionLengthMinutes AS bingeSessionLengthMinutes,
17   collect(DISTINCT content.title) AS consumedContent,
18   prefPlatform.name AS preferredPlatform;

```

platform	algorithmId	behavioralImpactId	audienceSegmentId	audienceLabel	bingeSessionLengthMinu	consumedContent	preferredPlatform
"Netflix"	"NetflixMainRecommendationAlgorithm"	"ViewingHabitChangeImpact"	"IsraeliYoungAdults18to34Group"	"Israeli Young Adults 18-34"	null	[]	null

Figure D.8: Perplexity algorithmic personalization query and audience cluster output.

Model 3 (ChatGPT Cypher):

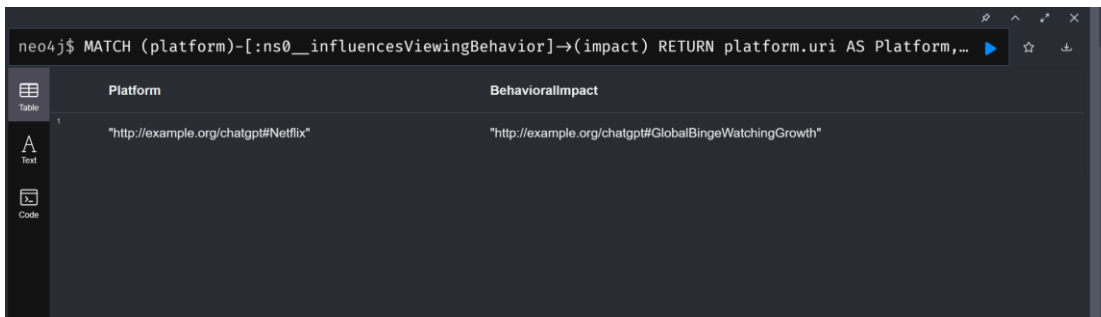
```

MATCH (platform)-[:ns0__influencesViewingBehavior]->(impact)

RETURN platform.uri AS Platform,

       impact.uri AS BehavioralImpact;

```



The screenshot shows a Neo4j Cypher query in a terminal window. The query is: `MATCH (platform)-[:ns0__influencesViewingBehavior]->(impact) RETURN platform.uri AS Platform, impact.uri AS BehavioralImpact;`. Below the query, the result is displayed in a table view with two columns: 'Platform' and 'BehavioralImpact'. The first row shows the platform URI as `"http://example.org/chatgpt#Netflix"` and the behavioral impact URI as `"http://example.org/chatgpt#GlobalBingeWatchingGrowth"`.

Platform	BehavioralImpact
"http://example.org/chatgpt#Netflix"	"http://example.org/chatgpt#GlobalBingeWatchingGrowth"

Figure D.9: ChatGPT platform-level behavioral influence query result.

Appendix E: Free-Text AI Prompts & Multi-Model Comparative Responses

Free-Text Prompt 1: Content-Driven Conflicts & Competition

Prompt:

"Based strictly on the Knowledge Graph domain, list all streaming platforms that produce content items which triggered diplomatic tensions or cultural conflicts, and specify which content items triggered which conflicts."

Model 1 (Gemini) Response:

"Based on the defined Knowledge Graph, the streaming platform Netflix produces the local original series Fauda (Fauda_Series), which directly triggered the diplomatic and political conflict Diplomatic Row Fauda (Diplomatic_Row_Fauda). Additionally, Netflix causes the decline of movie theaters (Movie_Theater_Decline), which triggers the cultural conflict Scorsese Cinema Debate (Scorsese_Cinema_Debate) opposing the platform."

Model 2 (Perplexity) Response:

"There are no streaming platforms in your current graph that produce content items which triggered diplomatic tensions or cultural conflicts. What your graph does show (related but not answering the question) Netflix distributes Squid Game and operates in the United States market. EU AVMSD Policy responds to Homogenization impact and applies to Netflix. SoftPower geopolitical impact is directed at the United States. Kan 11 (traditional outlet) competes with Netflix. But none of these establish that a specific content item triggered a specific diplomatic tension or cultural conflict".

Model 3 (ChatGPT) Response:

"Based strictly on the Knowledge Graph, no specific content item is explicitly modeled as triggering a diplomatic tension or cultural conflict. However, the graph does identify direct industry competition: Netflix competes with DisneyPlus and BBCOne. Netflix also distributes SquidGame, StrangerThings, and Fauda, but no relationship in the graph establishes that these content items directly triggered a conflict".

Free-Text Prompt 2: Geopolitical Regulation & Content Restrictions

Prompt:

"Based strictly on the Knowledge Graph domain, which geographic regions enforce censorship policies that restrict specific streaming content, which content items are blocked, and which platforms produced them?"

Model 1 (Gemini) Response:

"According to the Knowledge Graph, the Middle East Region (Middle_East_Region) enforces the policy Turkish State Censorship (Turkish_State_Censorship). This policy restricts the content item Fauda (Fauda_Series), which is produced and distributed globally by Netflix."

Model 2 (Perplexity) Response:

"There are no geographic regions in your current graph that enforce censorship policies restricting specific streaming content, and therefore no content items are shown as blocked, nor platforms identified as producers of blocked content. What your graph does show (related but not answering the question) EU AVMSD Policy applies to Netflix and responds to Homogenization cultural impact. Israel Broadcasting Regulation originates from the United States (as modeled in your data). Netflix distributes Squid Game. But none of these establish that a specific country/region blocks or restricts a specific content item via a censorship policy"

Model 3 (ChatGPT) Response:

"Based strictly on the Knowledge Graph, the EuropeanContentQuota policy applies to Netflix, and Netflix is regulated by the EuropeanUnionMediaRegulator. The graph also specifies that the EuropeanContentQuota requires local content from Netflix. However, it does not explicitly model a geographic region enforcing censorship of a specific content item, nor does it identify any individual streaming content item as blocked."

Free-Text Prompt 3: Behavioral Shifts & Demographic Audience Impact

Prompt:

"Based strictly on the Knowledge Graph domain, identify the demographic groups that experience socio-behavioral impacts (such as sleep loss or isolation) caused by global streaming platforms, and mention any correlated secondary impacts."

Model 1 (Gemini) Response:

"Based on the Knowledge Graph structure, the demographic cohort Youth Cohort (Ages 18–25) (Youth_Cohort_18_25) experiences the primary socio-behavioral impact Binge Sleep Loss (Binge_Sleep_Loss) caused by Netflix, as well as Youth Reading Drop (Youth_Reading_Drop). Furthermore, Binge Sleep Loss correlates directly with the secondary personal impact Teen Isolation (Teen_Isolation)."

Model 2 (Perplexity) Response:

"Only one demographic group is explicitly linked to socio-behavioral impacts in your graph: Heavy Binge Watchers (BingeWatcherProfile) consume Squid Game content. Global Middle Class (SocioEconomicGroup) is affected by Homogenization cultural impact and prefers Netflix. Israeli Young Adults 18-34 (AgeCohort) are the target audience for Haaretz Print and are personalized by Netflix's recommendation algorithm, which shapes Viewing Habit Change behavioral impact. However, your graph does not explicitly model sleep loss, isolation, or other specific socio-behavioral impacts beyond the generic ViewingHabitChangeImpact. There are no relationships connecting audience segments directly to impacts like "sleep loss" or "isolation.""

Model 3 (ChatGPT) Response:

“Based strictly on the Knowledge Graph, Netflix influences the behavioral impact GlobalBingeWatchingGrowth. However, the graph does not explicitly connect a particular demographic or audience group to this behavioral impact, nor does it model specific effects such as sleep loss or isolation or any correlated secondary behavioral impacts.”

Appendix F: Wikidata SPARQL Queries, Execution Links & Results

Wikidata SPARQL Query 1: Non-US Productions Distributed by Global Platforms

General Research Goal: Identify international television series originating outside the United States that are distributed globally by Netflix (Squid Game, Money Heist, Lupin, Elite).

Direct Query Link: [Run Query 1 on Wikidata Query Service](#)

SPARQL Code:

```
SELECT DISTINCT ?seriesLabel ?countryLabel ?languageLabel ?distributorLabel
WHERE {

  VALUES ?series { wd:Q106582931 wd:Q29647346 wd:Q104082429 wd:Q53000974 }

  ?series wdt:P495 ?country ;

    wdt:P364 ?language ;

    wdt:P750 ?distributor .

  FILTER(?distributor = wd:Q907311)

  SERVICE wikibase:label { bd:serviceParam wikibase:language "en,he". }
```

distributorLabel	languageLabel	countryLabel	seriesLabel
Netflix	Spanish	Spain	Money Heist
Netflix	French	France	Lupin
Netflix	Spanish	Spain	Elite
Netflix	Korean	South Korea	Squid Game

Figure F.1: International Netflix series metadata retrieved from Wikidata Query Service.

Wikidata SPARQL Query 2: European Union Media Regulations & Directives

General Research Goal: Retrieve formal European Union media directives and digital market regulations governing online platforms (such as the Audiovisual Media Services Directive and GDPR).

Direct Query Link: [Run Query 2 on Wikidata Query Service](#)

SPARQL Code:

```
SELECT ?directive ?directiveLabel WHERE {  
  
  VALUES ?directive { wd:Q56856283 wd:Q1172506 }  
  
  SERVICE wikibase:label { bd:serviceParam wikibase:language  
    "en,he". }  
  
}
```

directiveLabel	directive
Audiovisual Media Services Directive 2010	wd:Q56856283 Q
General Data Protection Regulation	wd:Q1172506 Q

Figure F.2: European Union media and privacy regulatory entities retrieved from Wikidata.

Wikidata SPARQL Query 3: Digital Media Behavioral & Health Taxonomies

General Research Goal: Map behavioral concepts and physiological conditions related to digital media consumption (such as binge-watching and sleep disorders).

Direct Query Link: [Run Query 3 on Wikidata Query Service](#)

SPARQL Code:

```
SELECT ?item ?itemLabel WHERE {  
  
  VALUES ?itemLabel {"binge-watching"@en "sleep disorder"@en}  
  
  ?item rdfs:label ?itemLabel.  
  
  FILTER(LANG(?itemLabel) = "en")  
  
}
```

itemLabel	item
sleep disorder	wd:Q177190 Q
binge-watching	wd:Q15094181 Q

Figure F.3: Standardized health and behavioral taxonomy mapping for binge-watching and sleep disorders.